Multimodal Data Management

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Course: Algorithms, Data Structures and Databases

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1. Introduction and Context

This project was developed as part of the Algorithms, Data Structures and Databases course and serves as a practical introduction to the design and implementation of complete data management systems. The main objective is to explore how core data engineering concepts—such as structured data handling, efficient storage, automation, and reproducibility—can be integrated into a cohesive, end-to-end pipeline using modern tools and best practices.

The project focuses on a multimodal music data management scenario, where heterogeneous data sources related to musical tracks are processed together within a unified system. These modalities include structured metadata (song titles, artists, albums), textual information (descriptions and semantic tags), and audio-related features. By constructing this pipeline, students gain hands-on experience with the fundamental components of real-world data projects: ingestion, transformation, validation, and exploitation.

Beyond its technical scope, the project introduces DataOps principles, emphasizing automation, data quality control, traceability, and reproducibility. Through the integration of object storage (MinIO), Python-based processing, and a vector database (ChromaDB), the project provides an educational yet realistic environment for managing multimodal musical data and enabling AI-driven exploration and recommendation.

2. Project Objectives

The primary goal of this project is to design and implement a complete multimodal data management pipeline for a musical catalog and advisory system. The specific objectives are:

- **Design a multimodal data management system for music data.** The pipeline integrates musical metadata, textual descriptors, and audio-related features within a unified structure, enabling semantic relationships between different representations of musical content.

- **Implement a zonal DataOps-inspired pipeline architecture.** The system is organized into four zones—Landing, Formatted, Trusted, and Exploitation—each representing a different level of data maturity. Automated scripts control data transitions between zones, ensuring reproducibility, traceability, and consistent data quality.

These objectives reflect both the theoretical foundations of the course and their practical application to a realistic multimedia data scenario.

3. Data Context and Sources

Domain

The project is situated within the domain of **digital music catalogs and recommendation systems**, an area where data engineering and machine learning techniques are widely used to organize, analyze, and retrieve large collections of musical content. Modern music platforms rely on heterogeneous data sources, including metadata, user-generated text, and audio descriptors, to power search, discovery, and recommendation functionalities.

This domain is inherently multimodal. Textual and structured information captures descriptive and contextual aspects of music, while audio-related features encode acoustic and stylistic properties. The project therefore serves as an educational case study in managing and integrating these modalities under a unified data architecture.

Problem Statement

A common challenge in music data systems is the fragmentation of information across different formats and sources. Musical metadata, descriptive text, and audio characteristics are often stored separately, limiting the ability to perform integrated search or analysis.

The problem addressed by this project is the lack of a **structured and automated pipeline** capable of ingesting, processing, validating, and relating multimodal music data in a reproducible way. Without such a pipeline, tasks such as semantic search, similarity analysis, or content-based recommendation become difficult to scale and maintain.

The project aims to address this challenge by designing a DataOps-oriented pipeline that unifies multiple music-related modalities and prepares them for analytical and exploratory tasks.

Business Value

Although developed for academic purposes, the system demonstrates concepts with strong real-world relevance. Efficient multimodal music data management is essential for streaming platforms, digital archives, and music recommendation services.
A functional multimodal pipeline can:

- Enable faster and more accurate music discovery.
- Support educational and research use cases in music information retrieval.
- Provide a scalable and auditable foundation for recommendation systems.

By demonstrating how structured pipelines can harmonize heterogeneous music data, the project highlights the practical value of DataOps methodologies in multimedia applications.

Data Modalities

The dataset used in this project includes several complementary modalities:

- **Structured Metadata:**
  Information such as track name, artist, album, genre, release year, and duration.
- **Textual Modality:**
  Descriptive text and semantic labels associated with tracks, representing stylistic or thematic aspects of music.
- **Audio-Related Features:**
  Numerical descriptors derived from audio analysis (e.g., danceability, energy, valence), which capture acoustic and musical characteristics.

Together, these modalities allow the system to represent musical tracks from multiple perspectives, forming the basis for multimodal retrieval and recommendation.

Data Sources

The data used in this project originates from public and educational music datasets:

- A historical music dataset derived from Spotify-related sources (via Kaggle), providing metadata and audio features for tracks released between 1950 and 2019.
- Supplementary curated samples used to demonstrate multimodal integration and processing.

This combination results in a representative yet manageable dataset suitable for educational experimentation.

4. Data Management Backbone

Zonal Architecture

The pipeline follows a zonal architecture inspired by DataOps, separating data according to processing stage and quality level. The four zones—Landing, Formatted, Trusted, and Exploitation—ensure transparency, reproducibility, and quality assurance throughout the data lifecycle.

Landing Zone

The Landing Zone acts as the entry point for all raw music data.

- Raw files and datasets are ingested without modification.
- Original formats and metadata are preserved.
- MinIO is used as object storage to maintain a single source of truth.

This zone ensures that all downstream processing can be reproduced from the original data.

Formatted Zone

In the Formatted Zone, data is standardized and normalized.

- Column names and data types are harmonized.
- Textual fields are cleaned and normalized.
- Audio feature fields are validated and cast to consistent numeric formats.

These transformations ensure compatibility across modalities and prepare the data for quality validation.

Trusted Zone

The Trusted Zone enforces data quality and consistency.

- Duplicate records are detected and removed.
- Missing or invalid values are handled according to predefined rules.
- Validation summaries and quality reports are generated automatically.

Only clean, reliable data is promoted to the exploitation stage.

Exploitation Zone Overview

The Exploitation Zone focuses on analytical usage.

- Textual and audio-derived representations are transformed into embeddings.
- Embeddings are stored in a vector database for efficient similarity search.
- Metadata is attached to each vector to support interpretability.

This zone enables semantic search, similarity analysis, and recommendation experiments.

Summary of Zones

Zone	Purpose	Tools / Techniques
Landing Zone	Raw data ingestion and secure storage	Python, MinIO
Formatted Zone	Format standardization and normalization	Pillow, text preprocessing

Trusted Zone	Data validation and anonymization	Python, logging
Exploitation Zone	Embedding generation and multimodal search	Sentence Transformers, ChromaDB

5. Implementation Summary

Tools and Environment

The project uses an open-source technology stack to simulate a realistic data engineering environment:

Component	Purpose	Technology
Storage	Object storage	MinIO
Processing	ETL and validation	Python, Pandas
Embeddings	Semantic representations	Sentence Transformers
Retrieval	Vector similarity search	ChromaDB
Automation	Orchestration	Docker, Python scripts

All components are containerized using Docker Compose, ensuring portability and reproducibility across environments.

Pipeline Execution

A centralized orchestration script executes all pipeline stages sequentially, enabling one-command execution of the full workflow. Configuration is managed via environment variables, separating infrastructure details from application logic.

6. Analytical Tasks

The pipeline supports several analytical tasks relevant to music data:

- **Similarity Search:** Identifying musically similar tracks based on combined textual and audio representations.
- **Cross-Modal Exploration:** Linking textual descriptions or tags with audio-based characteristics.
- **Catalog Exploration:** Supporting structured queries over genres, time periods, and musical styles.

These tasks demonstrate how multimodal data management enables richer interaction with music catalogs.

7. Data Quality Report (Trusted Zone)

The Trusted Zone produces a data quality report summarizing:

- Number of records processed and validated.
- Duplicates and inconsistencies removed.
- Coverage of audio and textual features.

This report ensures transparency and reproducibility of the data preparation process.

8. Operations and Automation

The pipeline is fully automated and reproducible:

- A single orchestration script runs all zones.
- Docker ensures consistent execution environments.
- Logs and reports provide auditability.

This operational design reflects DataOps best practices.

9. Results

Upon execution, the pipeline successfully processes the music dataset through all four zones. The data is standardized, validated, and indexed for semantic retrieval. The resulting system supports similarity search and exploratory analysis over a multimodal music catalog, confirming the effectiveness of the pipeline design.

10. Conclusions and Next Steps

This project demonstrates a functional **multimodal music data management pipeline** that integrates structured metadata, text, and audio-related features using DataOps principles. The system provides a solid educational foundation for building music recommendation and discovery tools.

Future work in Part 2 focuses on embedding evaluation, modality alignment, and performance analysis, extending the pipeline from data management into experimental multimodal modeling and assessment.